← Go to **ACL ARR 2026 May** homepage (/group?id=aclweb.org/ACL/ARR/2026/May)

# StructSynth: Dependency Graphs as Generation Plans for Low-Data Tabular Synthesis with Language Models

(/pdf?id=32AX5rq8ip)

*Siyi Liu (/profile?id=~Siyi_Liu1), Yujia Zheng (/profile?id=~Yujia_Zheng1), Haoyang LI (/profile?id=~Haoyang_LI3), Yongqi Zhang (/profile?id=~Yongqi_Zhang2)*

26 May 2026 (modified: 24 Jun 2026) ACL ARR 2026 May Submission
May, Senior Area Chairs, Area Chairs, Reviewers, Authors, Secondary Reviewers
Revisions (/revisions?id=32AX5rq8ip)


**Edit**

**Keywords:** Tabular Data Synthesis, Dependency Structure Discovery, Structure- Conditioned Generation, Data Augmentation, Low-Data Regimes

**Abstract:**
Tabular data derives its value from inter-feature dependencies, yet preserving them during synthesis is fragile when samples are scarce. Existing approaches either learn dependencies implicitly through distribution fitting, rely on statistical graph learning that becomes unstable with few samples, or encode structure through flat text serialization. Recent graph-aware methods use dependency graphs as attention biases or as backbones for non-LLM samplers, but they do not use the graph as a prompt-level plan for organizing the LLM's own generation process. We introduce StructSynth, a framework that treats a dependency graph as a generation plan---determining the generation order, conditioning context, and scope of each black-box LLM call. In Evidence-Grounded Graph Induction, LLM reasoning and statistical association cues jointly construct a Directed Acyclic Graph (DAG) from limited samples. In Graph-Planned Conditional Synthesis, this DAG drives autoregressive synthesis in topological order, conditioning each feature on previously generated values with graph-specified parent structure guiding each step, making the conditioning schedule explicit throughout synthesis. Experiments show that StructSynth achieves state-of-the-art downstream utility and privacy preservation while maintaining competitive statistical fidelity, especially in low-data scenarios.

**Paper Type:** Long
**Research Area:** NLP Applications
**Research Area Keywords:** financial/business NLP, security/privacy, NLP for social good
**Contribution Types:** NLP engineering experiment
**Languages Studied:** English
**Reassignment Request Area Chair:** This is not a resubmission
**Reassignment Request Reviewers:** This is not a resubmission
**A1 Limitations Section:** This paper has a limitations section.
**A2 Potential Risks:** Yes
**A2 Elaboration:** In Ethical Considerations section
**B Use Or Create Scientific Artifacts:** Yes
**B1 Cite Creators Of Artifacts:** Yes
**B1 Elaboration:** Section 4 and Appendix A cite the public datasets, baselines, and software libraries used in the experiments.
**B2 Discuss The License For Artifacts:** No
**B2 Elaboration:** We cite the artifacts used, but we do not provide a detailed license-by-license discussion. The artifacts are used only for research benchmarking.
**B3 Artifact Use Consistent With Intended Use:** Yes

**B3 Elaboration:** The Ethical Considerations section states that public datasets are used only as research benchmarks, and that generated records should not be used for high-stakes decision making without additional review and validation.
**B4 Data Contains Personally Identifying Info Or Offensive Content:** No
**B4 Elaboration:** We use publicly available benchmark datasets and do not collect new data. We discuss data sensitivity and privacy risks in Ethical Considerations, but we do not provide a separate PII/offensive-content audit.
**B5 Documentation Of Artifacts:** Yes
**B5 Elaboration:** Section 4 and Appendix "Detailed Experimental Setup" document the datasets, including domains, tasks, feature types, and evaluation setup.
**B6 Statistics For Data:** Yes
**B6 Elaboration:** Section 4 and Appendix "Detailed Experimental Setup" report dataset statistics, train/test splits, few-shot sample sizes, and random seeds.
**C Computational Experiments:** Yes
**C1 Model Size And Budget:** Yes
**C1 Elaboration:** Appendix "Implementation and Hyperparameters" reports model settings and downstream model parameters. Appendix "Efficiency Analysis" reports token usage for CLLM and StructSynth, including input/output tokens, total tokens, and relative overhead. Since the main LLMs are hosted/API models, we report token budget rather than GPU hours.
**C2 Experimental Setup And Hyperparameters:** Yes
**C2 Elaboration:** Section 4 and Appendix "Detailed Experimental Setup" describe the experimental setup, including datasets, baselines, train/test splits, sample sizes, seeds, and hyperparameter settings.
**C3 Descriptive Statistics:** Yes
**C3 Elaboration:** Section 4 reports results over multiple datasets and random seeds, including mean performance and variability/statistical comparisons where applicable.
**C4 Parameters For Packages:** Yes
**C4 Elaboration:** Appendix "Detailed Experimental Setup" reports implementation details and parameter settings for baselines, downstream models, LLMs, and evaluation components.
**D Human Subjects Including Annotators:** No
**D1 Instructions Given To Participants:** N/A
**D2 Recruitment And Payment:** N/A
**D3 Data Consent:** N/A
**D4 Ethics Review Board Approval:** N/A
**E Ai Assistants In Research Or Writing:** Yes
**E1 Information About Use Of Ai Assistants:** No
**E1 Elaboration:** We used AI assistants only for language polishing and editing support. All technical content, experiments, analyses, claims, and final writing decisions were reviewed and approved by the authors.
**Author Submission Checklist:** yes
**EMNLP 2026 AI Reviewing Experiment:** yes
**TLDR:** StructSynth learns a dataset's hidden structure and uses it as a blueprint to guide an LLM, generating more realistic synthetic data from scarce samples.
**Preprint:** no
**Preprint Status:** There is a non-anonymous preprint (URL specified in the next question).
**Existing Preprints:** https://arxiv.org/abs/2508.02601 (https://arxiv.org/abs/2508.02601)
**Preferred Venue:** EMNLP
**Visa Needs:** yes
**Consent To Share Data:** yes
**Consent To Share Submission Details:** On behalf of all authors, we agree to the terms above to share our submission details.
**Association For Computational Linguistics - Blind Submission License Agreement:** On behalf of all authors, I agree
**Submission Number:** 9229

## Discussion (?id=32AX5rq8ip#discussion)

Filter by reply type... Filter by author... Search keywords... Sort: Newest First

Everyone | Program Chairs | Submission9229... | Submission9229 Area... | Submission9229... | Submission9229 Authors | Submission9229... | Submission9229... | Submission9229... | Submission9229... | ✖

13 / 13 replies shown

Add: **Withdrawal** **Author-Editor Confidential Comment**

## General Response to All Reviewers

Official Comment

by Authors (Yongqi Zhang (/profile?id=~Yongqi_Zhang2), Haoyang LI (/profile?id=~Haoyang_LI3), Yujia Zheng (/profile?id=~Yujia_Zheng1), Siyi Liu (/profile?id=~Siyi_Liu1))

14 Jul 2026, 01:37

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers, Reviewers Submitted, Authors

**Comment:**

We thank all reviewers for the constructive feedback. The reviews raise five families of concerns: (a) novelty relative to PAFT, (b) reliance on semantic priors, (c) evaluation choices, (d) artifact availability, and (e) the calibration of privacy and fidelity claims.

| # | Addresses | New evidence | Headline result | Where addressed |
|---|---|---|---|---|
| Exp-1 | (a) | Bidirectional StructSynth×PAFT component transfer | The full StructSynth system outperforms the full PAFT system on utility and privacy (AUC 0.830 vs. 0.752); the StructSynth graph improves both generators (+0.030/+0.022 AUC) | 7Wh2-W1 |
| Exp-2 | (b) | Task-and-field anonymization | Functional under full anonymization — classification within 1.5 AUC points (0.865 → 0.850), fidelity improves; regression degrades more ($R^2$ 0.560 → 0.462) | 7Wh2-W4; iRH3-W3 |
| Exp-3 | (c) | Post-cutoff vary-n (Anxiety, Salary) | StructSynth leads at every n on both datasets; largest margin +3.31 AUC at n = 20 | q74j-W2 |
| Exp-4 | (c) | Qwen2.5-32B backbone check | AUC 0.856 vs. 0.843; $R^2$ 0.532 vs. 0.518 (0–1 scale) | q74j-W2 |

(d) is addressed in our response to Reviewer q74j (Artifact Checklist); (e) requires wording revisions only — no new experiments (details in our responses to 7Wh2, W2/W3).

Together, these results support that StructSynth's improvements arise from the explicit graph-as-generation-plan design rather than LLM memorization, hold across post-cutoff datasets and LLM backbones, and complement rather than duplicate PAFT's fine-tuning approach.

---

**Revision plan.** We will (1) position PAFT in §1–§2 [7Wh2-W1]; (2) sharpen the contribution statement to inference-time graph execution and include PAFT as a baseline in the main comparison [7Wh2-W1]; (3) correct GraDe in Appendix G.2 and justify baselines in §4.1 [7Wh2-W1; q74j-W1]; (4) replace "privacy preservation" wording with empirical-privacy language in §4.2/Limitations [7Wh2-W2]; (5) report the utility–privacy–fidelity trade-off in §5/Appendix K.2 [7Wh2-W3]; (6) add Exp-1–4, surface the direction-aware SHD evaluation in §4.4 [iRH3-W1], and clarify low-data/opaque-schema boundaries in §4.5/Appendix [iRH3-W2/W3; 7Wh2-W4; q74j-W2]; and (7) move the generation-plan definition earlier, expand the Appendix A artifact checklist, and identify direct structural metrics as future work [q74j-Minor/Checklist; 7Wh2-W1/W4].

Add: **Author-Editor Confidential Comment**

## Official Review of Submission9229 by Reviewer 7Wh2

Official Review by Reviewer 7Wh2 06 Jul 2026, 09:53 (modified: 12 Jul 2026, 23:35)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer 7Wh2

Revisions (/revisions?id=6SiIlYQEKB)

**Paper Summary:**
This paper focuses on tabular data synthesis using large language models (LLMs). The idea is to first induce a dependency DAG from a small real table using both LLM reasoning and statistical association measures, then use that DAG as a generation plan for black-box LLM calls. Basically the graph is used to determine feature order and prompt conditioning context.

**Summary Of Strengths:**

- The paper does a nice job and is akin to Bayesian network papers that separate structure discovery from conditional generation.
- The empirical results are strong. The authors demonstrate that their approach has the best downstream performacne on the datasets considered.
- The ablations are helpful. Replacing the LLM-guided graph discovery, or removing association scores, or ignoring topological order all hurt performance.

**Summary Of Weaknesses:**

- My main concern with this paper is that it is quite close to the paper by S. Xu et al. "Why LLMs Are Bad at Synthetic Table Generation and what to do about it" (https://arxiv.org/abs/2406.14541 (https://arxiv.org/abs/2406.14541)). This paper argues that LLM tabular generation fails because autoregressive generation imposes an order over columns, and that random feature-order fine-tuning will violate functional dependencies. It proposes injecting knowledge of functional relationships. Just like the proposed paper, there is a DAG learning step and then using that DAG information to generate data. As a result, the novelty here is limited and further the experimental results need to be contrasted with this above paper and their PAFT algorithm (in fact they seem to use the same dataset(s) as well).
- Besides the key related work contrasting and comparison, the privacy evaluation feels underdeveloped and is not quite a contribution.
- The statistical fidelity results are mixed (the algorithm proposed appears middle of the pack in some of these results).
- The limitations section acknowledges that opaque schemas such as V1/V2 or genomic locus identifiers weaken the LLM semantic prior (whereas this may not be a problem for the PAFT paper which seems to be discovering the dependencies).

**Comments Suggestions And Typos:**
Please see above.

**Confidence:** 5 = Positive that my evaluation is correct. I read the paper very carefully and am familiar with related work.
**Soundness:** 3 = Acceptable: This study provides sufficient support for its main claims. Some minor points may need extra support or details.
**Excitement:** 1.5
**Overall Assessment:** 2 = Resubmit next cycle: I think this paper needs substantial revisions that can be completed by the next ARR cycle.
**Ethical Concerns:**
There are no concerns with this submission.

**Needs Ethics Review:** No
**Reproducibility:** 4 = They could mostly reproduce the results, but there may be some variation because of sample variance or minor variations in their interpretation of the protocol or method.
**Datasets:** 4 = Useful: I would recommend the new datasets to other researchers or developers for their ongoing work.
**Software:** 4 = Useful: I would recommend the new software to other researchers or developers for their ongoing work.
**Knowledge Of Or Educated Guess At Author Identity:** No
**Knowledge Of Paper:** N/A, I do not know anything about the paper from outside sources
**Knowledge Of Paper Source:** N/A, I do not know anything about the paper from outside sources
**Impact Of Knowledge Of Paper:** N/A, I do not know anything about the paper from outside sources
**Reviewer Certification:** I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to

improving grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.
**Publication Ethics Policy Compliance:** I did not use any generative AI tools for this review

Add: **Author-Editor Confidential Comment** **Review Issue Report**

# New Review Issue Report

* denotes a required field

## I1 Not Specific

Example: missing references are not specified.

☐ The review is not specific enough.

## I2 Reviewer Heuristics

Examples: 'not SOTA', 'not novel', 'not suprising', 'too simple'. Note that such criticisms *may* be legitimate, if the reviewer explains their reasoning and backs it up with arguments/evidence/references. This flag should only be used to report cases where the reviewer has not done due diligence.

☐ The review exhibits one or more of the reviewer heuristics discussed in the ARR reviewer guidelines: https://aclrollingreview.org/reviewertutorial

## I3 Score Mismatch

'Overall assessment' score is a combination of several factors. Two of them are reflected in the 'soundness' and 'reproducibility' scores, which assess the technical merit of the submission. If they are low, this should be backed up by the text of the review. The 'excitement' score is subjective and may not be justified in text (e.g. different researchers have different ideas of what is exciting and presentation-worthy).

☐ The review score(s) do not match the text of the review.

## I4 Unprofessional Tone

Examples: rude reviews, sexist/racist/ageist etc. insinuations.

☐ The tone of the review does not conform to professional conduct standards.

## I5 Expertise

Examples: reviews that are not based on a deep understanding of the submission, or the core methodology used in this research area. This rubric can also be used for reviews suspected of being auto-generated.

☐ The review does not evince expertise.

## I6 Type Mismatch

Example: a short paper expected to provide more experiments than is necessary to support the stated claim.

☐ The review does not match the type of paper.

## I7 Contribution Mismatch

Examples: experimental results expected from a paper relying on a different methodology.

☐ The review does not match the type of contribution.

## I8 Missing Review

Example: one-liner reviews with generic criticisms.

☐ The review is missing or is uninformative.

## I9 Late Review

Example: the review came too late to be addressed in the author response.

☐ The review was late.

### I10 Unreasonable Requests

Example: requests for comparisons with the latest 'closed' models when it is not relevant for the research question.

- [ ] The reviewer requests experiments that are not needed to demonstrate the stated claim.

### I11 Non Response

The reviewers are volunteers, and are not required to respond to all author comments. Many do not respond in the forum, but do edit their reviews after seeing the author response. You should only use this rubric when there is a critical misunderstanding or unnoticed evidence, which would significantly impact key claims made in the review.

- [ ] The review does not acknowledge critical evidence in the author response.

### I12 Revisions Unacknowledged

[For revised submissions only:] the review does not acknowledge the revisions documented in revision notes, without sufficient justification.

- [ ] The review does not acknowledge the revisions

### I13 Other

Please explain your issue in sufficient detail below.

- [ ] Some other technical violation of the peer review process.

### Justification *

Describe the issue(s) with this review, clearly and concisely, with supporting evidence. You can use markdown. Please start the description for each type of issue with a new paragraph that starts with the review issue code. For example: `I2. The reviewer states [...]. We believe that this corresponds to review issue type I2, because [...].`
In case of reviewers not changing their scores based on the discussion, it is not in your interest to try to present a one-sided view of a reasonable scientific disagreement. Please include the link to the specific comment.

**Write** **Preview**

TeX is supported (https://docs.openreview.net/reference/openreview-tex)

**Readers ***

ACL ARR 2026 May Program Chairs

ACL ARR 2026 May Submission9229 Senior Area Chairs

ACL ARR 2026 May Submission9229 Area Chairs

*signatures*

**Signatures** * *signatures*

**Edit History**

**Readers** * *readers*

**Signatures** * ACL ARR 2026 May Submission9229 Authors

Submit Cancel

## Official Comment by Authors

Official Comment

by Authors ( Yongqi Zhang (/profile?id=~Yongqi_Zhang2), Haoyang LI (/profile?id=~Haoyang_LI3), Yujia Zheng (/profile?id=~Yujia_Zheng1), Siyi Liu (/profile?id=~Siyi_Liu1))

16 Jul 2026, 18:09

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer 7Wh2

**Comment:**

Dear Reviewer 7Wh2,

Thank you again for your thorough review. We wanted to follow up to ensure you had a chance to see our detailed response, which includes new experiments (Exp-1, Exp-2) directly addressing your concerns on novelty relative to PAFT, privacy evaluation, fidelity trade-offs, and opaque schemas. We would greatly appreciate any further feedback during the remaining discussion period and are happy to run additional analyses if needed.

Best regards, The Authors

Add: Author-Editor Confidential Comment

## Response to Reviewer 7Wh2 (1/2)

Official Comment

by Authors ( Yongqi Zhang (/profile?id=~Yongqi_Zhang2), Haoyang LI (/profile?id=~Haoyang_LI3), Yujia Zheng (/profile?id=~Yujia_Zheng1), Siyi Liu (/profile?id=~Siyi_Liu1))

14 Jul 2026, 01:29

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer 7Wh2

**Comment:**

We thank Reviewer 7Wh2 for the careful and thorough review. The central question is whether StructSynth is redundant given PAFT. We show it is not, at three levels: mechanism (inference-time execution vs. fine-tuning compilation), family-level evidence already in the submission (GraDe, Table 1), and a new direct comparison (Exp-1); Exp-2 additionally probes the opaque-schema boundary (W4).

Experiment numbering (Exp-1–4) is shared across all responses and defined in the General Response.

## W1: Relationship to PAFT and Novelty

> "My main concern with this paper is that it is quite close to the paper by S. Xu et al. 'Why LLMs Are Bad at Synthetic Table Generation and what to do about it' [...] Just like the proposed paper, there is a DAG learning step and then using that DAG information to generate data. As a result, the novelty here is limited and further the experimental results need to be contrasted with this above paper and their PAFT algorithm [...]"

**Both methods are structure-aware, but the structure plays a different operational role.** PAFT mines functional dependencies and compiles them into a global feature order for LoRA fine-tuning; StructSynth builds a DAG from semantic and statistical evidence and executes it at inference to control generation order, conditioning context, and prompt scope for a black-box LLM, without parameter updates (§3.1–§3.2; Eq. 4).

We agree both methods follow a "discover structure, then generate" template, and PAFT should have been cited — we will position it in §1–§2. The template, however, is where the operational overlap ends: this distinction is most consequential in our target low-data regime, where semantic priors complement weak association estimates and StructSynth achieves the lowest or near-lowest SHD at $n \leq 50$ (§4.4).

The family-level contrast the review asks for is already in the submission: GraDe is, like PAFT, an FD-discovery-plus-FD-guided generator trained by fine-tuning, and under identical 100-shot splits and the same evaluation pipeline StructSynth leads it on all six datasets (average utility 75.01 vs. 51.05; Table 1). To our knowledge, no prior method executes a discovered dependency graph as the generation plan for a black-box LLM; structure-aware predecessors (PAFT, GraDe) inject structure at fine-tuning time.

Exp-1 now extends the comparison to PAFT itself: we evaluated bidirectional component transfer at 100 real rows and 1,000 synthetic rows with a unified XGBoost evaluator.

| Generator | Structural source | Avg. AUC | Avg. $R^2$ | Fidelity error ↓ | DCR deviation ↓ |
|---|---|---|---|---|---|
| PAFT | PAFT FD graph | 0.752 | 0.360 | 0.537 | 0.154 |
| PAFT | StructSynth dependency graph | 0.782 | 0.404 | 0.540 | 0.150 |
| StructSynth | PAFT FD graph | 0.808 | 0.550 | 0.580 | 0.053 |
| StructSynth | StructSynth dependency graph | 0.830 | 0.591 | 0.595 | 0.032 |

**In this descriptive comparison, the full StructSynth system leads the full PAFT system on every utility and privacy column (AUC 0.830 vs. 0.752; $R^2$ 0.591 vs. 0.360; DCR deviation 0.032 vs. 0.154).** The component-transfer rows isolate the contribution of the structural source: holding the generator fixed, the StructSynth dependency graph consistently improves downstream utility over the PAFT FD graph. Under the PAFT generator it raises classification AUC by +0.030 (0.752 → 0.782) and $R^2$ by +0.044; under the StructSynth executor the gains are similar (AUC 0.808 → 0.830; $R^2$ 0.550 → 0.591). Cross-compatibility is preserved in the reverse direction (a PAFT FD graph can still drive the StructSynth executor at AUC 0.808), confirming the two structural representations are interchangeable at the interface level while differing in downstream strength.

## W2: Scope of the Privacy Evaluation

> "Besides the key related work contrasting and comparison, the privacy evaluation feels underdeveloped and is not quite a contribution."

We agree DCR evidence alone should not be read as a standalone privacy contribution; the revision will replace the "privacy preservation" wording (abstract, §1, §4.2, §5) with empirical-privacy language.

**StructSynth pairs the best utility rank (1.00) with the best privacy rank (1.50), which we attribute to conditioning each generation call on a local subgraph (§4.2; Appendix K.2).**

DCR is not differential privacy; the regularization effect was not designed as a privacy mechanism; and the submitted Ethics section already states that StructSynth offers no formal privacy guarantee (Ethics, l.574–578). Appendix K.2 discusses the resulting fidelity–privacy trade-off.

Add: Author-Editor Confidential Comment

## Response to Reviewer 7Wh2 (2/2)

Official Comment

by Authors ( Yongqi Zhang (/profile?id=~Yongqi_Zhang2), Haoyang LI (/profile?id=~Haoyang_LI3), Yujia Zheng (/profile?id=~Yujia_Zheng1), Siyi Liu (/profile?id=~Siyi_Liu1))

14 Jul 2026, 01:31

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer 7Wh2

**Comment:**

# W3: Statistical Fidelity Trade-off

> "The statistical fidelity results are mixed (the algorithm proposed appears middle of the pack in some of these results)."

The reviewer is correct about the mixed ranking.

**Pairwise fidelity and downstream utility are different objectives: the Bayesian-Sampler ablation attains the lowest fidelity error (48.86) yet the worst utility (81.17), while StructSynth holds utility rank 1.00 — the objective for which augmentation is deployed (Table 3).**

Fidelity error measures pairwise-correlation matching, whereas the generation graph encodes conditional parent–child dependencies (§4.2, l.356–359 and l.413–417; Appendix K.2). The same tension appears among the real baselines, where high fidelity often stems from memorization: BN ranks 3.00 on fidelity error but 9.00 on privacy, and GReaT ranks 4.00 on fidelity error with the worst privacy risk (90.15%) (§4.2, l.396–404; Table 2). The submitted profile is utility rank 1.00, privacy rank 1.50, fidelity rank 8.50; the revision will report this trade-off explicitly in §5, and we do not claim comprehensive superiority.

# W4: Opaque and Anonymized Schemas

> "The limitations section acknowledges that opaque schemas such as V1/V2 or genomic locus identifiers weaken the LLM semantic prior (whereas this may not be a problem for the PAFT paper which seems to be discovering the dependencies)."

Opaque schemas do weaken the semantic prior — the submission flags exactly this boundary (Appendix K.3).

**Under full anonymization of names and descriptions, StructSynth stays functional — classification loses 1.5 AUC points while fidelity improves — because association scores keep the graph populated when semantics are absent (Exp-2).**

We anonymized all column names and task/column descriptions while keeping values unchanged (Exp-2) *(gpt-5-mini, 100 real rows, 1,000 synthetic rows; 0–1 scale)*.

| Dataset | Condition | Utility | Fidelity ↓ | DCR (ideal 0.50) |
|---|---|---|---|---|
| Anxiety | Original | AUC 0.865 | 0.579 | 0.447 |
| Anxiety | Anonymized | AUC 0.850 | 0.534 | 0.564 |
| Salary | Original | $R^2$ 0.560 | 0.646 | 0.501 |
| Salary | Anonymized | $R^2$ 0.462 | 0.615 | 0.412 |

Salary's larger regression drop ($R^2$ 0.560 → 0.462) shows semantic priors matter most for fine-grained continuous dependencies (full analysis in our response to Reviewer iRH3, W3). This limitation is mirrored on the PAFT side: FD discovery plus fine-tuning requires statistical signal and parameter access — the two resources scarcest in our target regime — whereas StructSynth runs against black-box APIs and its graph transfers across eight backbones (§4.6, Figure 5). The two approaches therefore compose rather than compete: when schemas are opaque, a statistically discovered FD graph plugs directly into StructSynth's executor — Exp-1's transfer cell (StructSynth generator + PAFT graph, AUC 0.808) already demonstrates this fallback.

---

**Revision plan.** We will: (1) cite/position PAFT §1–§2 (W1); (2) sharpen the contribution statement to inference-time graph execution, add Exp-1, and include PAFT as a baseline in the main comparison (W1); (3) correct GraDe description in Appendix G.2 (W1); (4) empirical-privacy wording in the abstract, §1, §4.2, §5, and Limitations (W2); (5) three-dimensional profile §5/Appendix K.2 (W3); (6) add Exp-2, opaque-schema boundary + statistical-graph fallback (W4); (7) direct structural metrics as future work (W1/W4).

We hope these results address your concerns — in particular the novelty assessment, in light of the two-step differentiation, the family-level GraDe comparison already in Table 1, and the direct PAFT study (Exp-1). We would be glad to run further analyses during the discussion period.

Add: **Author-Editor Confidential Comment**

## Official Review of Submission9229 by Reviewer q74j

Official Review by Reviewer q74j 29 Jun 2026, 17:05 (modified: 14 Jul 2026, 20:18)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer q74j

Revisions (/revisions?id=OY4fiQT2uE)

**Paper Summary:**
This paper proposes a method for generating additional tabular data in scarce data scenarios. They decouple the extraction task in which a DAG is extracted from the existing tabular data using an LLM, and the generation task, where additional data is generated using the extracted DAG as guidance.

**Summary Of Strengths:**
The paper is very well-written The experimental design is sound. The method is evaluated on six downstream tasks and the extracted graph is evaluated for its quality as well. They show with extensive experiments that method is suitable for the task at hand The ablations further strengthen the claims and show the contribution of the different elements in the design

**Summary Of Weaknesses:**
It is not clear whether the baselines used are the best available methods in their respective categories. As the authors explain in l.052-l.055, deep generative models' performance varies across architectures and some may require many more examples to train. I expected a little explanation in the experiments section that motivates the choice for the particular models used as baselines. It is not always clear why you select a particular dataset to show results. In Section 4.5 and 4.6 why did you choose the Adult dataset and not one of the datasets that are created after the knowledge cutoff for LLMs?

**Comments Suggestions And Typos:**
The goal of the paper only became fully clear to me in l. 197. This description could feature earlier in the introduction.

**Confidence:** 4 = Quite sure. I tried to check the important points carefully. It's unlikely, though conceivable, that I missed something that should affect my ratings.
**Soundness:** 4.5
**Excitement:** 3.5
**Overall Assessment:** 4 = Conference: I think this paper could be accepted to an *ACL conference.
**Limitations And Societal Impact:**
The limitation section is insightful and to the point. The paper also discusses ethical considerations well.

**Ethical Concerns:**
The authors describe ethical considerations well.

**Needs Ethics Review:** No

**Reproducibility:** 4 = They could mostly reproduce the results, but there may be some variation because of sample variance or minor variations in their interpretation of the protocol or method.
**Datasets:** 4 = Useful: I would recommend the new datasets to other researchers or developers for their ongoing work.
**Software:** 3 = Potentially useful: Someone might find the new software useful for their work.
**Knowledge Of Or Educated Guess At Author Identity:** No
**Knowledge Of Paper:** N/A, I do not know anything about the paper from outside sources
**Knowledge Of Paper Source:** N/A, I do not know anything about the paper from outside sources
**Impact Of Knowledge Of Paper:** N/A, I do not know anything about the paper from outside sources
**Reviewer Certification:** I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to improving grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.
**Publication Ethics Policy Compliance:** I did not use any generative AI tools for this review

Add: **Author-Editor Confidential Comment** **Review Issue Report**

− = ≡

## Response to Reviewer q74j

Official Comment
by Authors ( Yongqi Zhang (/profile?id=~Yongqi_Zhang2), Haoyang LI (/profile?id=~Haoyang_LI3), Yujia Zheng (/profile?id=~Yujia_Zheng1), Siyi Liu (/profile?id=~Siyi_Liu1))
14 Jul 2026, 01:35
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer q74j

**Comment:**
We thank Reviewer q74j for the careful and constructive review, and especially for recognizing the paper's writing, experimental design, and ablations. Below, we address each point in order. Experiment numbering (Exp-1–4) is shared across all responses and defined in the General Response.

# W1: Baseline Selection

> "It is not clear whether the baselines used are the best available methods in their respective categories. [...] I expected a little explanation in the experiments section that motivates the choice for the particular models used as baselines."

**We selected twelve baselines to cover three paradigms — DGMs, structure-aware, and LLM-based — with a foundational method and a recent representative in each, under the same low-data setting (§4.1, l.334–348).**

The suite covers DGMs (TVAE/CTGAN, TabDDPM, NFlow, TabSyn), structure-aware methods (BN, GOGGLE, DECAF, SPADA-NF), and LLM-based methods (GReaT, CLLM, GraDe). TabSyn (2024), SPADA-NF (2025), and GraDe (2025) ensure that every category includes a recent method, while the established methods provide recognizable anchors. Evaluating all methods at n = 100 is deliberate because data scarcity is the paper's target condition; Table 1 reports average scores of 63.69 for TabDDPM and 68.04 for TabSyn in this regime — large-sample reputation alone does not identify the best method under data scarcity.

# W2: Adult versus Post-Cutoff Datasets

> "In Section 4.5 and 4.6 why did you choose the Adult dataset and not one of the datasets that are created after the knowledge cutoff for LLMs?"

**Adult was chosen for comparability with prior work and consistency with the Table 3 ablation; we now complement it with Anxiety and Salary, both released after the LLM knowledge cutoff (Exp-3, Exp-4).**

The submission already identifies Anxiety and Salary as post-cutoff datasets (§4.1, l.321–325; §4.3, Table 3), while using Adult across the ablation, vary-n, and backbone analyses enables a controlled analytical thread. We compared CLLM and StructSynth on both post-cutoff datasets for n = 20, 50, 100, 200 (Exp-3):

*(gpt-5-mini, 1,000 synthetic rows per run)* (AUC / $R^2$ × 100)

| Dataset / Method | n = 20 | n = 50 | n = 100 | n = 200 |
|---|---|---|---|---|
| Anxiety AUC / CLLM | 77.52 | 81.87 | 85.17 | 85.44 |
| Anxiety AUC / StructSynth | **80.83** | **83.50** | **86.45** | **87.23** |
| Salary $R^2$ / CLLM | 48.16 | 52.34 | 54.53 | 61.39 |
| Salary $R^2$ / StructSynth | **50.56** | **54.90** | **55.98** | **63.54** |

StructSynth leads at every n on both datasets; the largest margin is +3.31 AUC at n = 20.

We also ran a Qwen2.5-32B backbone check on both post-cutoff datasets (Exp-4) *(n = 100, 1,000 synthetic rows per run; 0–1 scale)*:

| Dataset | Metric | CLLM | StructSynth |
|---|---|---|---|
| Anxiety | AUC | 0.843 ± 0.012 | **0.856 ± 0.014** |
| Salary | $R^2$ | 0.518 ± 0.037 | **0.532 ± 0.036** |

The advantage holds in the same direction on both post-cutoff datasets under Qwen2.5-32B, suggesting that the gain is not specific to the GPT backbone.

# Minor: Earlier Goal Description

> "The goal of the paper only became fully clear to me in l. 197. This description could feature earlier in the introduction."

We appreciate this presentation suggestion. The three-part definition of a generation plan (generation order, conditioning context, prompt scope) will move to the Introduction immediately after the research question at l.085–087.

# Artifact Checklist

> "Reproducibility: 1 [...] Datasets: 1 [...] Software: 1"

**The anonymous software artifact was already part of the original submission (Appendix A, l.855–859); the checklist below makes the remaining data-to-result path explicit.**

| Artifact | Status and evidence |
|---|---|
| Code and instructions | Anonymous repository and usage instructions included in the original submission (Appendix A, l.855–859). |
| Dataset provenance and protocol | All six datasets are public and cited; Appendix G.1 specifies the 8:2 split and seeds 42–51 (§4.1; Appendix G.1, Table 4). |
| Exact replication inputs | We will add preprocessed datasets, exact split files, and configurations. |
| Preprocessing and evaluation | We will add the preprocessing scripts and an artifact index mapping each reported result to its command/configuration. |

**Revision plan.** We will: (1) add paradigm-coverage and low-data criteria to §4.1 (W1); (2) explain Adult choice, add Exp-3 to §4.5 and Exp-4 to §4.6 (W2); (3) move generation-plan definition after l.085–087 (Minor); (4) expand Appendix A into an artifact checklist with replication assets (Checklist).

We hope these clarifications address your concerns, including the Reproducibility, Datasets, and Software assessments. We would be glad to provide any additional information during the discussion period.

Add: **Author-Editor Confidential Comment**

## response to author response

Official Comment by Reviewer q74j 14 Jul 2026, 20:22

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer q74j

**Comment:**
I thanks the authors for their explanations and additions. I would be happy for these to appear in the camera-ready version. I have changed the scores for reproducibility, dataset and software. I am sorry for not having noticed these contributions. I leave my other scores unchanged.

Add: **Author-Editor Confidential Comment**

*Replying to response to author response*

## Official Comment by Authors

Official Comment
by Authors ( Yongqi Zhang (/profile?id=~Yongqi_Zhang2), Haoyang LI (/profile?id=~Haoyang_LI3), Yujia Zheng (/profile?id=~Yujia_Zheng1), Siyi Liu (/profile?id=~Siyi_Liu1))

16 Jul 2026, 18:10

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer q74j

**Comment:**
Dear Reviewer q74j,

Thank you very much for reading our response and for the updated scores. We appreciate the constructive exchange and will incorporate everything in the camera-ready version.

Best regards, The Authors

Add: **Author-Editor Confidential Comment**

## Official Review of Submission9229 by Reviewer iRH3

Official Review by Reviewer iRH3 24 Jun 2026, 16:38 (modified: 18 Jul 2026, 15:07)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer iRH3

Revisions (/revisions?id=54FM9gJXkT)

**Paper Summary:**
This paper proposes StructSynth, a tabular data generation and augmentation method that first constructs a dependency graph among features using an LLM and then generates synthetic samples based on the learned graph structure. Learning dependencies among features is a central challenge in this setting and has been studied from various perspectives, including Bayesian networks, causal discovery, and statistical approaches. The key contribution of this work is to use the semantic knowledge of LLMs to infer the dependency graph, particularly in small sample size scenarios where purely data-driven approaches may struggle.

**Summary Of Strengths:**

1. The paper addresses an important problem in tabular data generation and augmentation, particularly under small sample size regimes where existing methods often face difficulties.
2. Experimental results are convincing, especially in terms of downstream model performance. The proposed method consistently performs well when the available sample size is small, suggesting that it may be practically useful across a range of applications.
3. The paper goes beyond reporting aggregate performance metrics and also presents examples of the learned dependency graphs. These qualitative results provide useful insights into the behavior of the method and help readers better understand the learned structures.

**Summary Of Weaknesses:**

1. Since the proposed method infers not only the existence of dependencies but also the directions of edges, the evaluation should assess more than statistical fidelity. While the paper reports measures such as

Statistical Fidelity Error, it would also be valuable to evaluate the correctness of the learned graph structure, particularly edge directions, on datasets where ground-truth dependency structures are available.
2. The method is specifically motivated by small sample size scenarios, and the experiments focus on sample sizes up to n = 200. However, it is important to understand how the method behaves as more data become available. Experiments with larger sample sizes would help clarify the trade-off between the advantages of LLM-based graph construction and purely data-driven alternatives.
3. As acknowledged in the limitations section, the proposed approach exploits semantic information contained in feature names and descriptions, whereas most non-LLM baselines cannot use such information. This raises questions about the fairness of the comparison. An informative additional experiment would be to anonymize feature names (e.g., replacing them with V1, V2, ...) and evaluate how much the performance depends on semantic information provided by the feature names.

**Comments Suggestions And Typos:**
Please address the weaknesses discussed above.

**Confidence:** 3 = Pretty sure, but there's a chance I missed something. Although I have a good feel for this area in general, I did not carefully check the paper's details, e.g., the math or experimental design.
**Soundness:** 3.5
**Excitement:** 3 = Interesting: I might mention some points of this paper to others and/or attend its presentation in a conference if there's time.
**Overall Assessment:** 3.5 = Borderline Conference
**Ethical Concerns:**
There are no concerns with this submission

**Needs Ethics Review:** No
**Reproducibility:** 4 = They could mostly reproduce the results, but there may be some variation because of sample variance or minor variations in their interpretation of the protocol or method.
**Datasets:** 3 = Potentially useful: Someone might find the new datasets useful for their work.
**Software:** 4 = Useful: I would recommend the new software to other researchers or developers for their ongoing work.
**Knowledge Of Or Educated Guess At Author Identity:** No
**Knowledge Of Paper:** N/A, I do not know anything about the paper from outside sources
**Knowledge Of Paper Source:** N/A, I do not know anything about the paper from outside sources
**Impact Of Knowledge Of Paper:** N/A, I do not know anything about the paper from outside sources
**Reviewer Certification:** I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to improving grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.
**Publication Ethics Policy Compliance:** I used a privacy-preserving tool exclusively for the use case(s) approved by PEC policy, such as language edits

Add: **Author-Editor Confidential Comment** **Review Issue Report**

## Official Comment by Authors

Official Comment
by Authors ( Yongqi Zhang (/profile?id=~Yongqi_Zhang2), Haoyang LI (/profile?id=~Haoyang_LI3), Yujia Zheng (/profile?id=~Yujia_Zheng1), Siyi Liu (/profile?id=~Siyi_Liu1))
16 Jul 2026, 18:10
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer iRH3

**Comment:**
Dear Reviewer iRH3,

Thank you for your constructive review. We wanted to gently follow up in case you had not yet had a chance to review our response, which addresses each of your concerns on direction-aware structural evaluation, behavior beyond the low-data regime, and the semantic-prior boundary with a new anonymization ablation (Exp-2). We would value any further comments or questions during the discussion period.

Best regards, The Authors

Add: **Author-Editor Confidential Comment**

## Response to Reviewer iRH3

Official Comment

by Authors ( Yongqi Zhang (/profile?id=~Yongqi_Zhang2), Haoyang LI (/profile?id=~Haoyang_LI3), Yujia Zheng (/profile?id=~Yujia_Zheng1), Siyi Liu (/profile?id=~Siyi_Liu1))

14 Jul 2026, 01:36

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer iRH3

**Comment:**

We thank Reviewer iRH3 for the careful and constructive review. We respond to each weakness in order, foregrounding (i) the direction-aware structural evaluation already in the submission (§4.4) and (ii) a new task-and-field anonymization ablation (Exp-2).

Experiment numbering (Exp-1–4) is shared across all responses and defined in the General Response.

# W1: Direction-Aware Structural Recovery

> "It would also be valuable to evaluate the correctness of the learned graph structure, particularly edge directions, on datasets where ground-truth dependency structures are available."

This is an important criterion that deserved greater visibility. **Section 4.4 already includes this evaluation: SHD against three ground-truth DAGs (Asia, Child, Insurance), where every reversed edge is explicitly penalized (§4.4, Figure 3, l.470–490).**

We evaluate Asia (8 nodes), Child (20), and Insurance (27) for $n \in \{20, 50, 100, 200\}$. Structural Hamming Distance counts the edge insertions, deletions, and reversals needed to recover the reference DAG; StructSynth attains the lowest or near-lowest SHD on all three, with its clearest advantage at $n \le 50$. Directionality is functionally necessary because the DAG defines the topological generation schedule, but it is not an ontological or causal claim (Appendix D). Consistent with this functional role, removing topological order alone lowers AUC by 1.1 points (Table 3). Replacing the StructSynth graph with PC- or NoTears-discovered alternatives — which differ in both edge set and orientation — lowers it by 1.0 and 1.4 points respectively.

# W2: Behavior Beyond the Low-Data Regime

> "Experiments with larger sample sizes would help clarify the trade-off between the advantages of LLM-based graph construction and purely data-driven alternatives."

**From $n = 20$ to 200, StructSynth remains stable across utility, fidelity, and privacy, with its largest advantage at $n \le 50$; the convergence trend the reviewer anticipates is already measurable in range (§4.5, Figure 4; §4.4, Figure 3).**

$n \le 200$ is the designed operating range (§1, l.040–047). The utility gap narrows toward $n = 200$, and PCI and NoTears approach StructSynth's SHD as n increases (§4.4, Figure 3) — exactly the trade-off the reviewer asks about. The paper already identifies the method as best suited to severely limited data (especially $n \le 100$) and notes diminished advantages when statistical structure learning becomes reliable (Appendix K.3).

These observations motivate the hypothesis that beyond $n = 200$, stronger association estimates will reduce the relative value of the LLM semantic prior while StructSynth's absolute performance remains competitive.

# W3: Task-and-Field Anonymization

> "An informative additional experiment would be to anonymize feature names [...] and evaluate how much the performance depends on semantic information provided by the feature names."

**StructSynth remains functional without semantic headers — statistical association scores compensate — with a task-dependent contribution from semantic priors (Exp-2).**

StructSynth pairs semantic reasoning with observed association scores (Cramér's V, |r|, and correlation ratio), so the statistical channel remains available when names are opaque (§3.1.2, l.249–258). We anonymized all column names and task/column descriptions while keeping values unchanged (Exp-2) *(gpt-5-mini, 100 real rows, 1,000 synthetic rows; 0–1 scale)*.

| Dataset | Condition | Utility | Fidelity ↓ | DCR (ideal 0.50) |
|---|---|---|---|---|
| Anxiety | Original | AUC 0.865 | 0.579 | 0.447 |
| Anxiety | Anonymized | AUC 0.850 | 0.534 | 0.564 |
| Salary | Original | $R^2$ 0.560 | 0.646 | 0.501 |
| Salary | Anonymized | $R^2$ 0.462 | 0.615 | 0.412 |

On Anxiety, anonymization costs 1.5 AUC points (0.865 → 0.850) but improves fidelity (0.579 → 0.534), suggesting the two channels play complementary roles: association-only graphs track pairwise statistics more tightly, while semantic edges add utility-relevant structure. On Salary, the original retains a larger advantage ($R^2$ 0.560 → 0.462), indicating that semantic priors contribute more to regression tasks involving fine-grained continuous dependencies.

**Revision plan.** We will: (1) rename §4.4 to surface SHD's edge-reversal penalty and direction-aware evaluation (W1); (2) state the operating range, convergence evidence, and clearly qualified n > 200 hypothesis in Limitations (W2); (3) add the anonymization ablation and semantic-prior boundary conditions to the appendix (W3).

We hope these results address your concerns; we would be glad to run further analyses during the discussion period.

Add: **Author-Editor Confidential Comment**

## Author-Editor Confidential Comment by Program Chairs

Author-Editor Confidential Comment by Program Chairs 13 Jun 2026, 14:41

Program Chairs, Senior Area Chairs, Area Chairs, Authors

**Comment:**
Dear authors,

In your submission you have some captions that are positioned above the tables instead of positioned below. Your submission will not be desk rejected for that, but please make sure to fix it according to the formatting guidelines at https://acl-org.github.io/ACLPUB/formatting.html (https://acl-org.github.io/ACLPUB/formatting.html) during re-submission or camera-ready stage.

Yours Sincerely, ARR Editors

Add: **Author-Editor Confidential Comment**